Lecture 31: ECL (Emitter Coupled Logic)

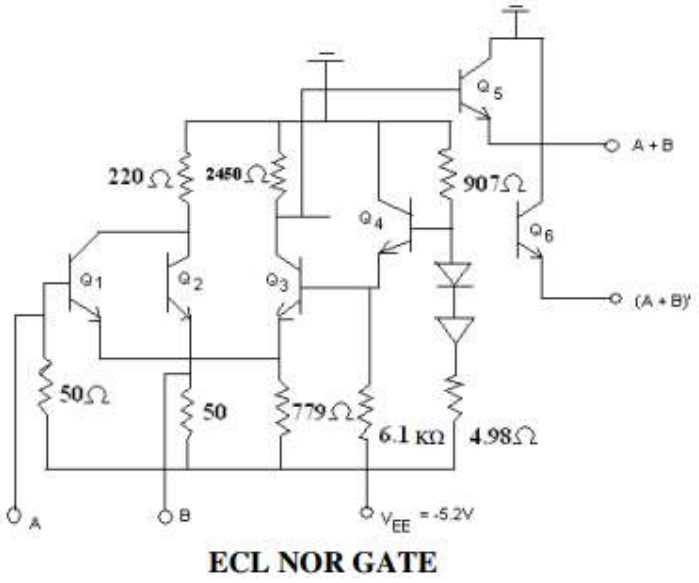
Emitter – Coupled Logic Families

ECL families use overdriven BJT differential amplifiers with single-ended inputs. The emitters of the transistors are connected all together, hence the name 'emitter-coupled' logic families.

The transistors in this system never saturate; their logic high and logic low levels are chosen close to one another, thus eliminating the possibility of saturation in the transistors. The transistors are thus operated in the active region or the cut-off region.

These transistors use the property of current switching. Thus this family is also known as Current Steering Logic (CSL) or Current Mode Logic Families (CML). It makes use of a transistor-based differential amplifier to amplify and combine digital signals.

Operation of ECL 2 input OR/NOR gate:



ECL NOR GATE

A basic two input ECL OR/NOR gate operation

The working is very obvious, if any input is high (-0.8V) then the corresponding transistor is turned on and Q_3 is turned off.

- This causes a voltage of -1.6V to appear at emitters of both of the transistors Q_1 & Q_2 Since V_{BB} = -1.3v_1 there is only a drop of 0.3V at V_{BE} of Q_5 , hence it is in cut–off. Current in the 220 Ω resistance flows through the conducting transistor and the output of Q_6 is low.

- Now if both inputs are low, only Q_3 will conduct, and Q_1 & Q_2 will be cut off this is because emitters will be at -2.1 V and each transistor has its base at -1.8V hence drop is only 0.3V hence they are in cut-off.

- In this situation current through the 220 Ω resistance flows out through Q_6 and O/P is high.

Truth table

A	B	NOR	OR
0	0	1	0
0	1	0	1
1	0	0	1
1	1	0	1

We can see here that the transistors are differential amplifiers, with their emitters coupled. Cryptographic applications and high-speed requirements use ECL Logic.

Characteristics of ECL Logic Families

- The output of an ECL system has a very low impedance, thus having a fan-out of about 25**.**
- The propagation time is about 1ns, making it the fastest logic family.
- A logic low voltage for ECL is about -1.7V to -1.75V**.**
- A logic high voltage for ECL is about -0.8V**.**
- It is the fastest amongst all the logic families.
- It has an average propagation delay time of about 1ns-4ns.

Advantages of ECL Logic Families

- Has a fan-out better than the TTL Logic family
- Offers the highest speed in operation
- ECL systems produce complementary outputs (OR-NOR, AND-NAND).
- Parameters do not vary much with temperature

Disadvantages of ECL Logic Families

- Worst noise immunity compared to TTL and CMOS
- Highest power consumption, when compared with TTL and CMOS
- VLSI design difficult as ECL circuits require resistors, thus increasing system size
- Capacitive loading reduces the fan-out capacity

Complimentary MOSFETs Logic Families

The CMOS family uses MOSFETs in the integrated circuits. Both NMOS and PMOS complement each other and are used symmetrically in each configuration of the circuit, hence the name ‘complimentary.’

Since both PMOS and NMOS are present, an application of input voltage will *turn on any one type* of the transistor at a time. Thus, at any time, there is no direct link between the power supply and the ground.

Two main attributes of CMOS devices are high immunity to noise and low static power consumption. CMOS consists of only enhancement type of MOSFETs.

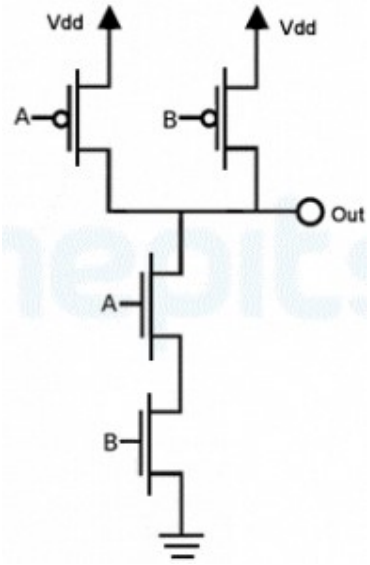


Fig: basic 2 input CMOS NAND gate

For NMOS transistors, if the input is a 1, the MOS turns on; otherwise, it is off. On the other hand, for the PMOS, if the input is 0, the MOSFET switches on; otherwise, the transistor is off. Thus, if either of the input is 1, the output voltage is equal to VDD.

The FETs are placed such that if either of the NMOS or PMOS transistors are connected in series, the other type is connected in parallel. Thus, only one type of MOS turns on at a time.

We use CMOS chips in RF applications, satellite communication, Bluetooth, and mobile networks. We also see their use in radar applications and Wi-Fi technology. To be honest, CMOS is the backbone of the semiconductor industry. Its applications are countless.

Characteristics of CMOS Logic Families

- CMOS supports a very large fan-out, more than 50 transistors.
- It has excellent noise immunity amongst all families.
- A logic low voltage for CMOS is about 0V to 0.05V.
- A logic high voltage for CMOS is somewhere between **4.5V to 5V**.
- The propagation delay is the worst when compared with TTL and ECL families at about 200ns.

Advantages of CMOS Logic Families

- Has the highest fan-out, when compared with TTL and ECL
- Works well over a wide range of temperature
- Noise immunity is better than TTL and ECL
- Least static power dissipation

Disadvantages of CMOS Logic Families

- Average propagation delay is the least in comparison with TTL and ECL

Common CMOS Logic ICs

- **4000** family
- **74C** family
- **74HC** family
- **74HCT** family
- **74AC** family
- **74ACT** family
- **74AHC** family

----------END OF LECTURE----------

REFERENCES

1. **Text book**: **T1.** Modern Digital Electronics by R. P. Jain (Tata McGraw Hill)
2**. Reference book**: **R1.** Fundamental of digital circuits by A. Anand Kumar (PHI)